

Population organization and closed-loop attractors for obstacle avoidance under perturbation

Summary

Motor systems must adjust actions after unexpected forces while maintaining goal-directed movement within physical boundaries. In human reaching, such forces can prompt graded corrections supported by coordinated muscle activity while the hand continues toward its goal. How can a learned controller organize these corrections when obstacles narrow the available route? We asked whether reward-based learning alone can produce structured obstacle avoidance and whether its population activity offers an organizational comparison with biological motor control. We trained a proximal policy optimization (PPO) actor-critic for 3 million environment steps. Its feedforward multilayer perceptron (MLP), with two 256-unit hidden layers, mapped each state to an action that moved a point mass through a corridor with two obstacles. We analyzed activity from a 256-unit hidden layer using principal component analysis (PCA), clustering, kinematic tuning, linear decoding, and closed-loop analysis of the controller coupled to the environment physics. Moderate lateral perturbations changed the path by less than 0.1 m without stopping progress toward the target. The first two principal components captured 71% of hidden-unit activity variance, and activity followed phase-ordered trajectories shared across perturbation conditions but separated by detour route. At the obstacle band, analysis of the policy and physics identified a stable lateral fixed point for each route ($x^*=3.32$ m and 6.71 m), with similar relaxation times (1.42 s and 1.34 s) and clearances (0.43 m and 0.46 m). These findings suggest reward training can produce organized, locally stabilizing routes in this model. Motor-cortex studies describe reaching-related activity at the population level, and human perturbation studies report graded corrections during movement. This is a limited organizational comparison, not evidence that model units reproduce neural or muscle mechanisms. Because results come from one training run, further seeds and comparisons with biological movement will be needed to assess robustness and relevance.

Additional details

To understand how the controller produced its routes, we examined both its hidden activity and the movement that activity helped generate. Here, “population” means the 256 artificial units in one hidden layer, not neurons recorded from an animal. The actor is feedforward and has no recurrent memory: it maps the current observation to an action. The fixed points therefore describe the policy acting together with the point-mass physics, not equilibria of the hidden layer alone. This differs from recurrent-network fixed-point analyses (Sussillo & Barak, 2013). Motor-cortex studies provide a useful organizational comparison because reaching-related activity is distributed across neural populations (Churchland et al., 2012), but artificial units are not neurons.

The controller was trained for 3 million environment steps. We evaluated an unperturbed condition and six lateral-force conditions: three impulse magnitudes in each direction, each lasting 14 timesteps. We used principal component analysis, clustering, kinematic tuning, and linear decoding to examine activity, then analyzed the closed loop to ask whether its routes had stable lateral states. The first two principal components captured 71% of hidden-unit activity variance. Activity followed phase-ordered paths shared across perturbation conditions but separated by detour route.

From population activity to movement

At the obstacle band ($y = 5$ m), with the force off and forward speed held at its plateau, we examined commanded lateral acceleration as position changed. A zero crossing with a negative slope means small sideways deviations elicit commands back toward the route. Figure 1 shows the trajectories and this stability analysis.

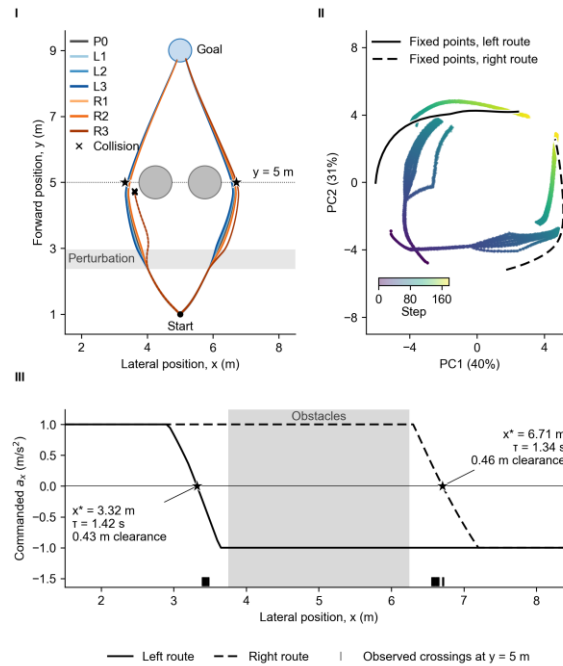


Figure 1. Task and closed-loop results. (A) Baseline and perturbed trajectories; only the strongest rightward impulse caused collisions, on the left route. Stars mark fixed points at $y = 5$ m. (B) Hidden activity in PC1-PC2 space, colored by movement step. (C) Commanded lateral acceleration; each route has a stable zero crossing with a negative slope.

The two fixed points were at $x = 3.32$ m and $x = 6.71$ m, within 0.11 m and 0.09 m of the successful episode crossings. Their routes left 0.43 m and 0.46 m of obstacle clearance. Local relaxation times were 1.42 s and 1.34 s; these describe the model returning toward its fixed points, not the latency of a human response. We next asked whether the hidden activity also carried information about which force had occurred.

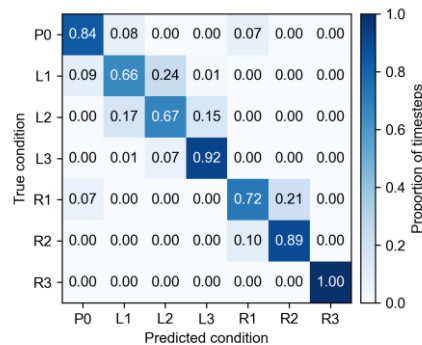


Figure 2. Prediction of perturbation condition from hidden activity after the impulse. Rows are true conditions and columns are predicted conditions; values are row-normalized proportions from validation that held out entire episodes (accuracy = 0.81; chance = 0.14). Errors are mostly between neighboring force magnitudes on the same side.

The decoder could read perturbation information after the impulse, but this does not show that the policy used it to choose an action. In biological reaching, unexpected forces also prompt graded corrections with coordinated muscle activity while the hand continues toward its goal (Pruszynski & Scott, 2012; Nashed et al., 2014). We see a modest organizational parallel, not evidence that the model reproduces neural or muscular mechanisms. These results come from one training run, and fixed-point locations vary across checkpoints. Further seeds and direct comparisons with human movement will be needed to assess how robust or informative this account is.